

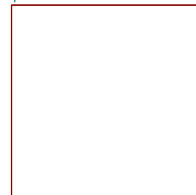


bare_metal

**FPGA dev board
iCE40UP5k**

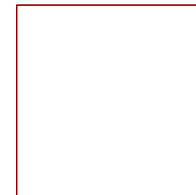
2026 by wenzellabs.de

pmod



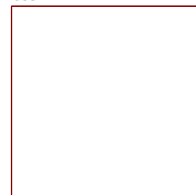
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flash



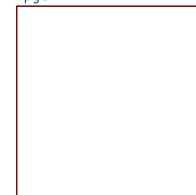
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usb



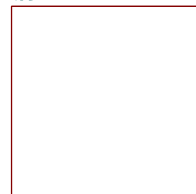
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fpga



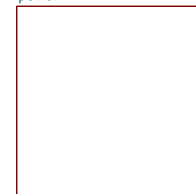
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led



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power



File: power.kicad_sch

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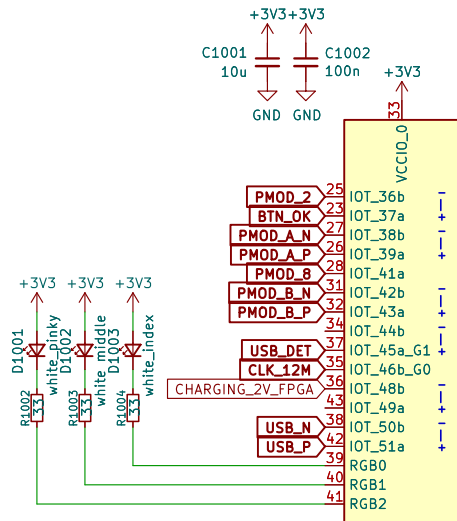
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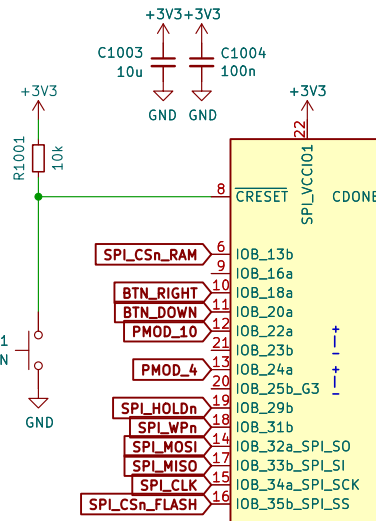
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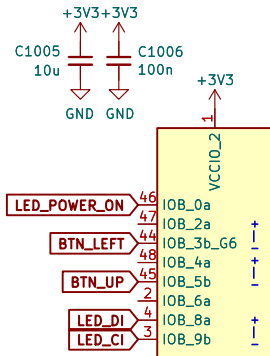
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U1001A
ICE40UP5K-SG48ITR



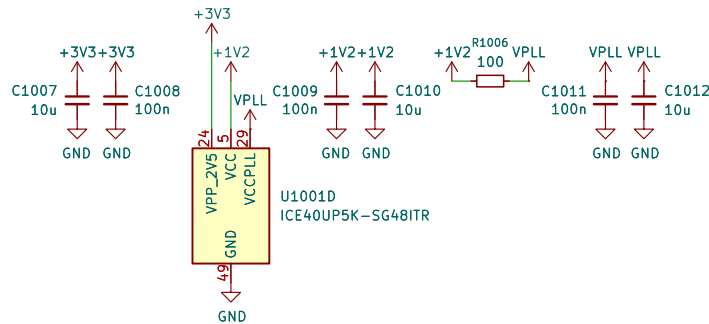
U1001B
ICE40UP5K-SG48ITR



U1001C
ICE40UP5K-SG48ITR

differential pairs:

+ sig_P
- sig_N



U1001D
ICE40UP5K-SG48ITR

Table 2.2. Recommended Power Filtering Groups and Components

Power Input	Recommended Filter	Notes
V _{CC}	4.7 μ F + 100 nF per pin	Core logic. 1.2 V
V _{CCIOx}	4.7 μ F + 100 nF per pin	I/O banks power supply pin V _{CCIOx} Banks 0, 1, 2 1.8 V, 3.3 V
VPP_2.5 V	4.7 μ F + 100 nF per pin	NVCM programming and operating supply voltage 2.5 V
SPI_VCC	4.7 μ F + 100 nF per pin	SPI supply voltage 1.8 V, 3.3 V
V _{CCPLL, 2}	100 Ω + 4.7 μ F + 100 nF per pin	PLL analog supply voltage 1.2 V

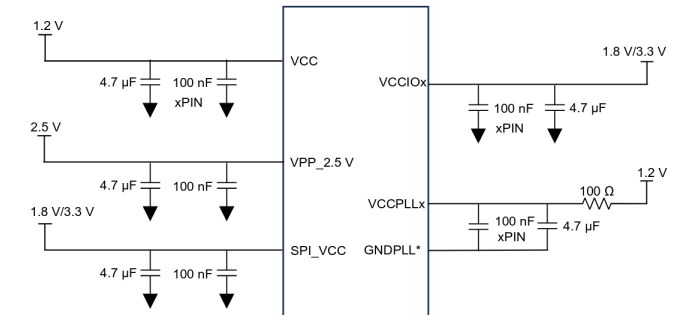
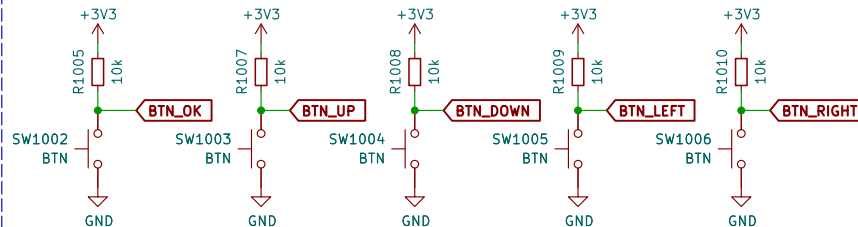
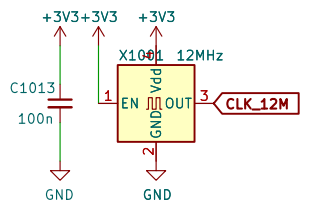


Figure 2.1. Typical Power Supply Filter

Clock: 12MHz osc

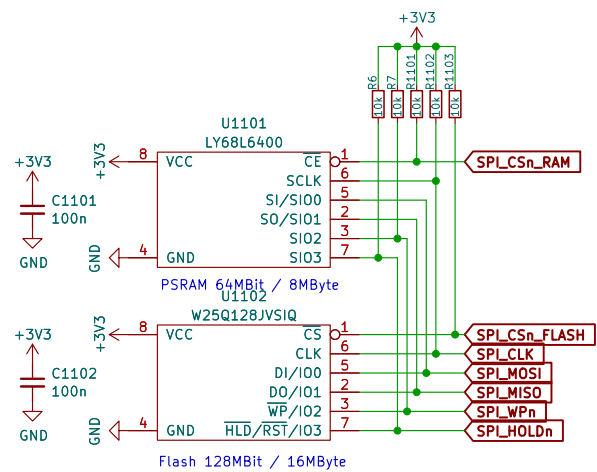


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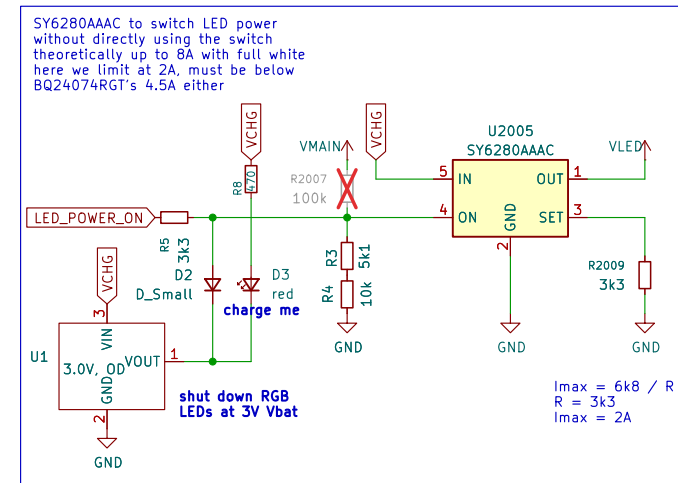
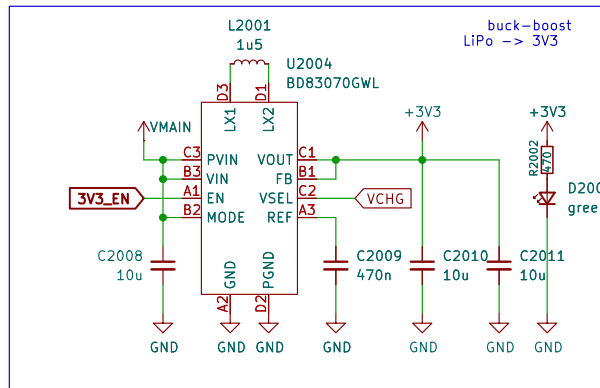
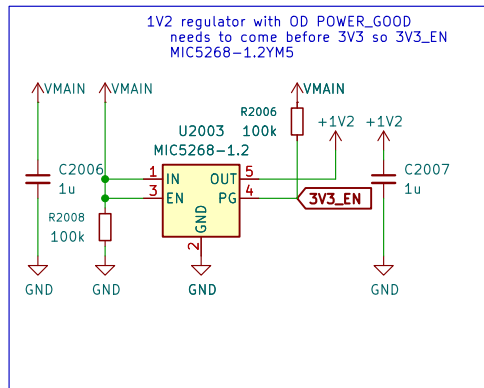
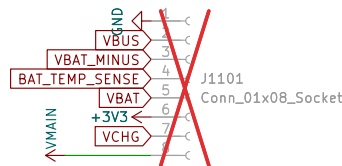
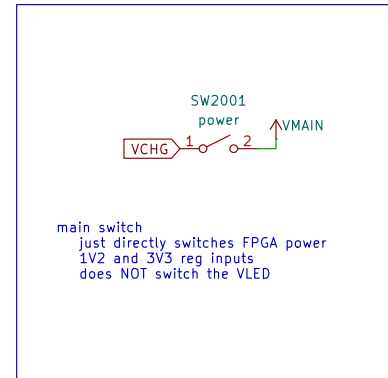
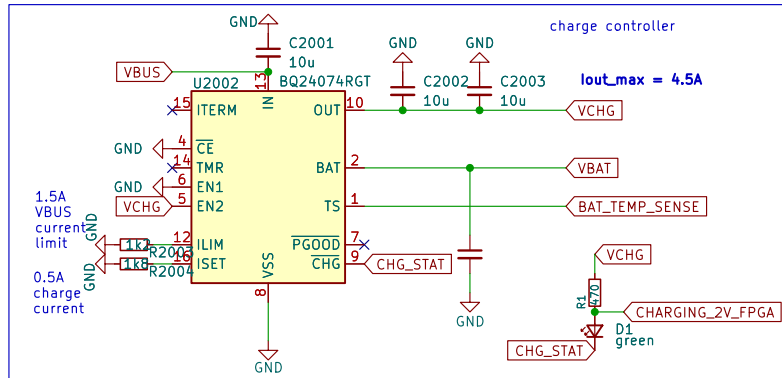
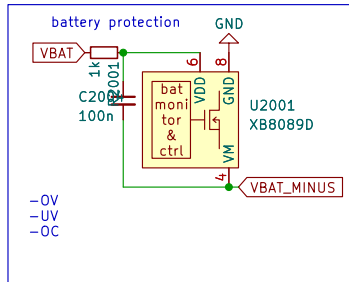
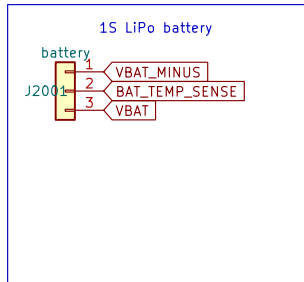
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Transmission Line Type

☐ Microstrip Line
☐ Coplanar wave guide
☐ Coplanar wave guide w/ ground plane
☐ Rectangular Waveguide
☐ Coaxial Line
☒ Coupled Microstrip Line
☐ Stripline
☐ Twisted Pair

Substrate Parameters

er: 4.6
tan δ: 0.011
p: 1.72e-08 Ω · m
H: 0.2104 mm
H_L: 1e+20 mm
T: 0.035 mm
Roughness: 0 mm
μconductor: 1

Physical Parameters

W: 0.338 mm
S: 0.18 mm
L: 100.0 mm

Analyze
Synthesize

Electrical Parameters

Z_{even}: 62.6615 Ω
Z_{odd}: 44.9822 Ω
Ang_L: 1.84814 rad

Results

Effective er (even): 3.63621
Effective er (odd): 3.13247
Unit propagation delay (even): 63.7754 ps/c
Unit propagation delay (odd): 59.1933 ps/c
Conductor losses (even): 0.145928 dB
Conductor losses (odd): 0.20319 dB
Dielectric losses (even): 0.0848857 d
Dielectric losses (odd): 0.0739888 d
Skin depth: 3.01275 μm
Differential Impedance (Z_d): 90.0066 Ω

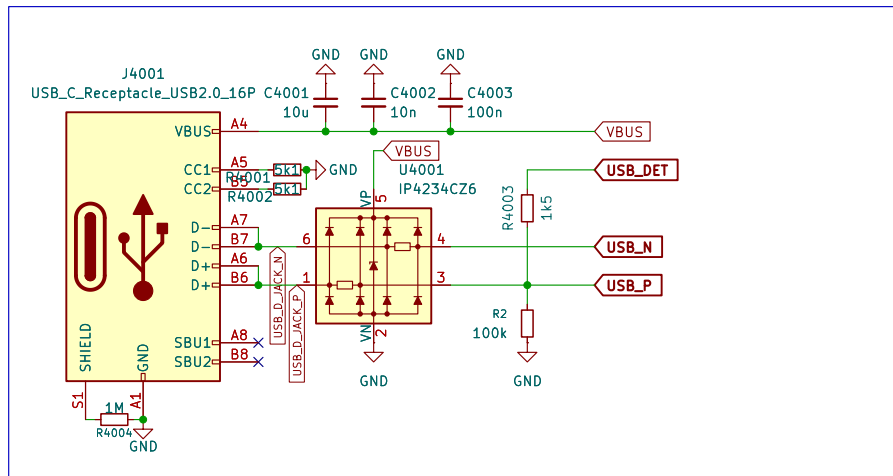
Component Parameters

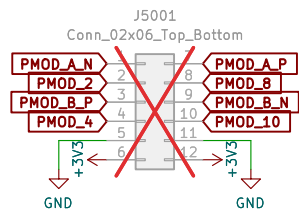
Frequency: 480 MHz

Odd

Even

USB differential impedance is 90R
 $Z_{diff} = 2 * Z_{odd}$





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